

A decorative graphic on the left side of the slide consisting of two overlapping parallelograms. The front one is blue and the back one is a light green. They are positioned diagonally, with the blue one partially covering the green one.

The Blank OSL

Presented by: Team Vertex (Alan, Evan, Pouria, Shirley, Tony)



Team Vertex

- Team of 5 members
- All on campus this term and next term
- Diverse expertise and background (SE + AI)
- Shared vision of building AI-driven application

Team organization

- 2 Weekly standup (30 minutes)
- Projects Tracker & Codebase: Github
- Product and Research Team
- Weekly check-in with mentor

The screenshot shows a Kanban board with three columns: **Todo** (2/10 items, Estimate: 0), **In Progress** (6/5 items, Estimate: 0), and **Done** (11 items, Estimate: 0). Each column has a description and a list of tasks.

- Todo:**
 - TheBlankOSL #1: Design a testing framework.
 - TheBlankOSL_UI #10: UI terminal can not send EOF input.
- In Progress:**
 - The_Blank_OSL_Research #4: Find more papers on dataset, eval, benchmark/testing.
 - TheBlankOSL_LLM_Client #3: Fine tune AI models.
 - TheBlankOSL_LLM_Client #2: System prompt engineering.
 - TheBlankOSL_UI #1: UI terminal sometimes output unprintable character.
- Done:**
 - TheBlankOSL_UI #7: Capture filesystem hierarchy from the home directory.
 - TheBlankOSL_LLM_Client #4: Write a parser to parse system property info that's passed through program main()'s arguments.
 - TheBlankOSL_UI #9: Capture operating system and user information.
 - The_Blank_OSL_Research #2: Read survey paper: The rise and potential of

2 Events

TEXT CHANNELS

general



api-team

ui-team

useful-link

meeting-notes

to-do



Advisor / Mentor

Ramani T.

Research Engineer

@ Google DeepMind



Evolution of User Interface

1970s - 1980s

Text-based user interface



1980s - 2007s

Graphical user interface



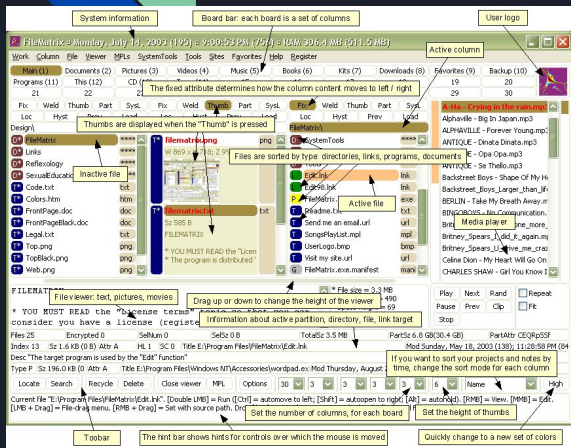
2007s - Present

Finger/Voice-based user interface





Our vision



Stupidly simple.



Stupidly intuitive.

How can you make money as a kid?

Google:



Bing:

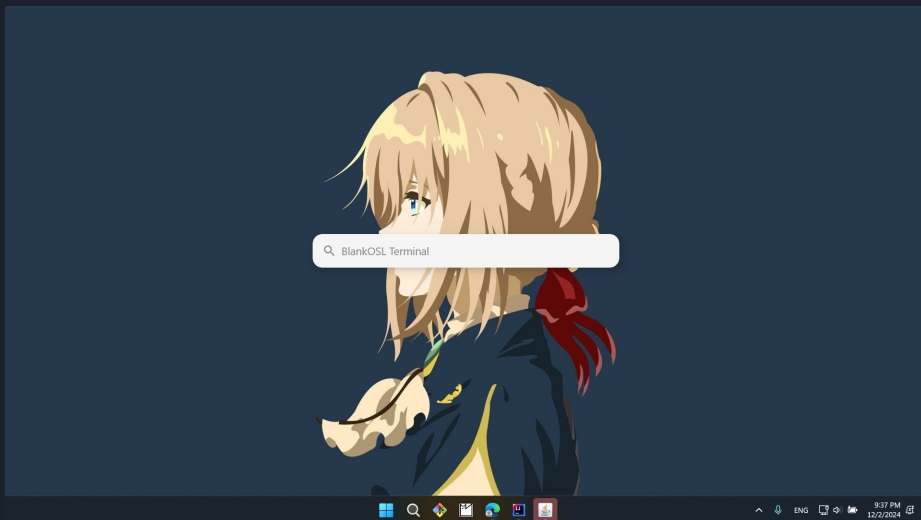
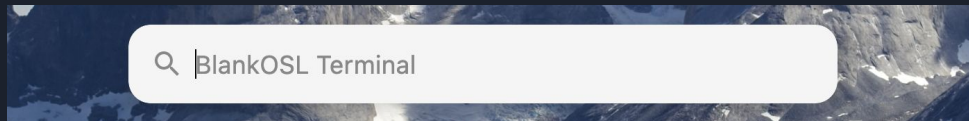


Stupidly smart.

Project Overview

The Blank OSL:

An Operating System Layer (OSL) utilizes LLMs to improve OS navigation.

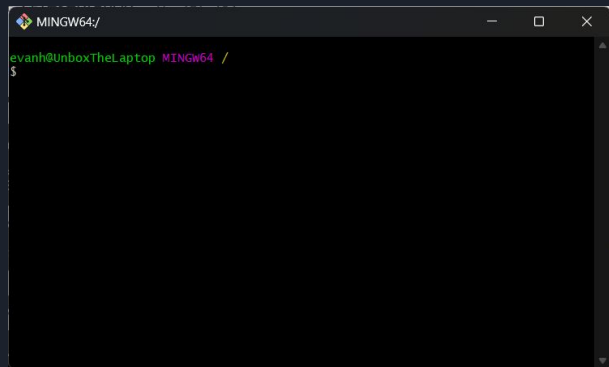


- The UI of all UI.
- Minimalist UI design.
- Beginner friendly with minimum knowledge required.
- Fast-paced environment that requires quick solution.

Current State

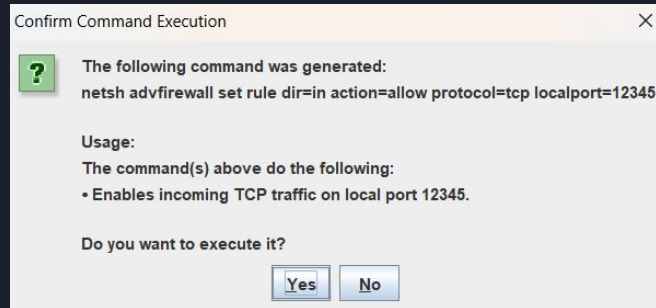
Terminal UI (Kotlin)

open port 12345 for inbound TCP



Abstract Command Executor

LLM Clients (Python)



Confirmation Dialog

Compatible in....



Linux



MAC OS



Windows



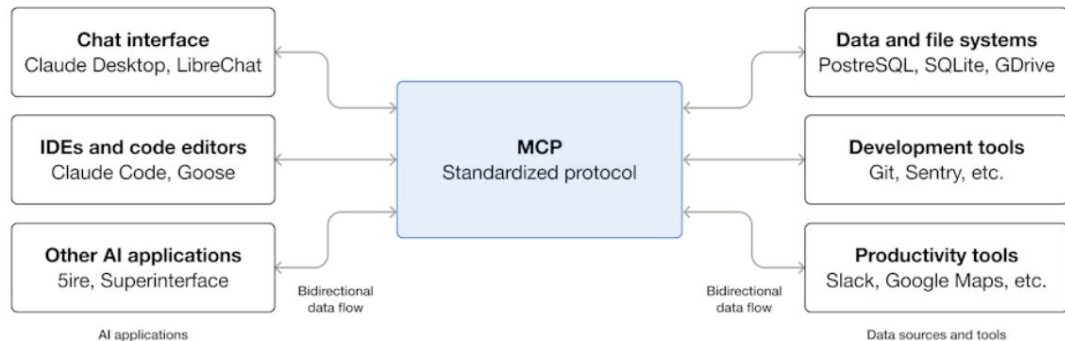
Use Cases

- Create a new folder named Capstone inside Documents.
- Delete the folder called temp in my home directory.
- Plot a math curve directly in the terminal window.
- Set alarm clocks

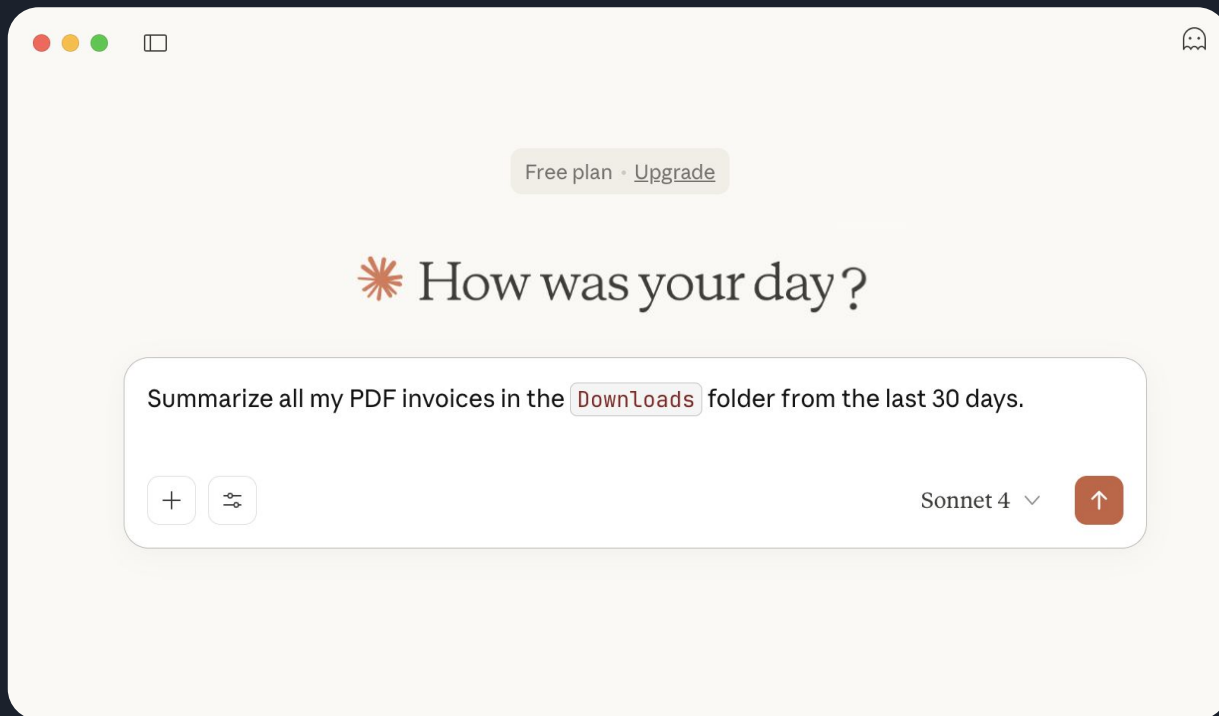
Do more with your terminal!

Pivot to Model Context Protocol (MCP)

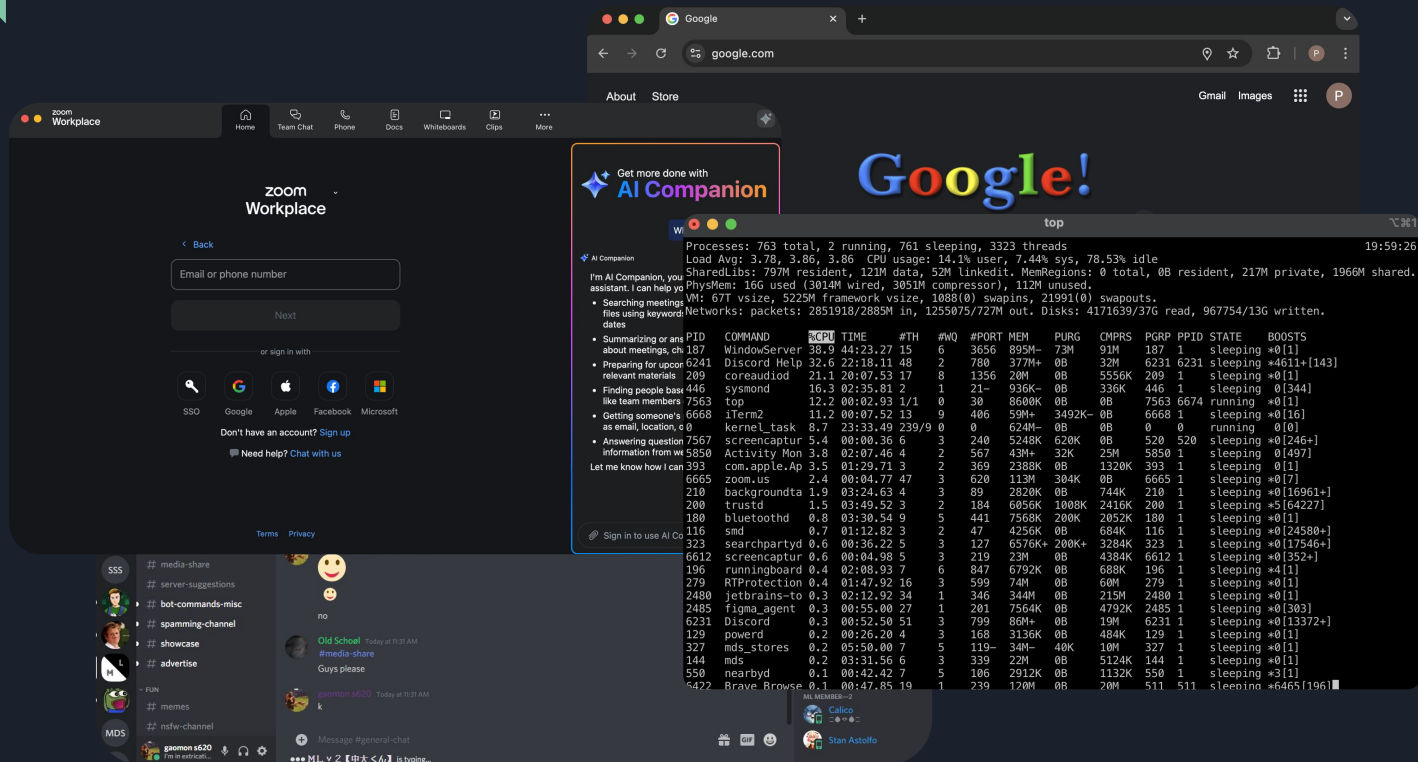
Think of MCP like a USB-C port for AI applications. MCP provides a standardized way to connect AI applications to external systems.



MCP Server for the user's device

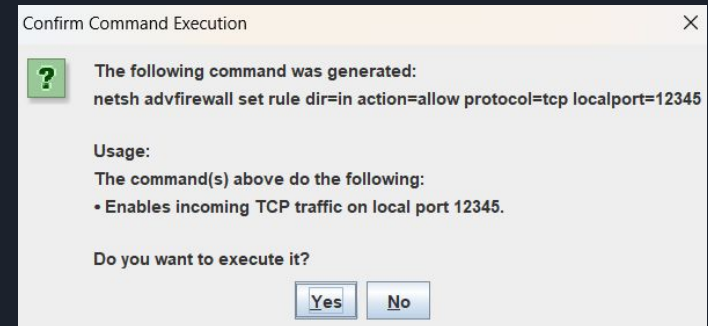
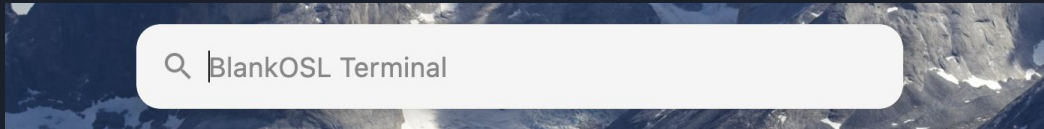


Automate Application



Future Plans

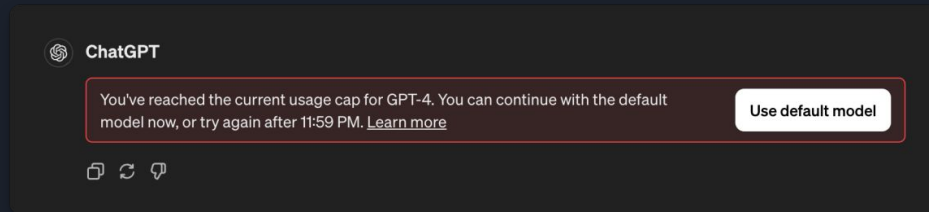
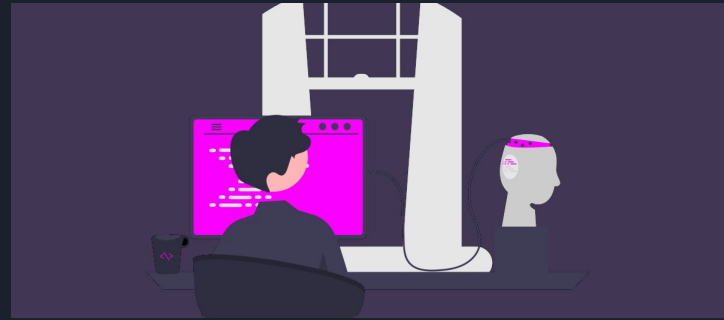
- Create MCP server and client (late October)
- User Confirmation Feature (late November)
- Speech to Text Feature (end of term)





Problems

- Running out of tokens/prompts
- Security vulnerability





Thank you!

Questions?